HAND IN

answers record on exam paper

Queens University Final Examination
Smith Engineering
Department of Electrical And Computer Engineering

ELEC 477 Distributed Systems
T.R. Dean
April 23, 2024

Student Number ______________________________

Student Number in words ____
(Example: two three two nine seven four two five)

INSTRUCTIONS TO STUDENTS:

- This examination is 3 HOURS in length.
- Answer all questions in the space provided on these pages. If you need extra space, use the extra pages at the end of the exam.
- **No aids are allowed.**
- This exam has 14 pages
- Total xx marks. Please pay attention to the number of marks for each question.
- The numbers in the bottom right of the page show a running total of the marks in the exam (8/30 on page 5 means 8 marks on page 5 and 30 marks for all pages up to and including page 5)

PLEASE NOTE:
Proctors are unable to respond to queries about the interpretation of exam questions. Do your best to answer exam questions as written.

1. **(6 marks)** Give a brief (one or two line) description of each of the following terms as they were discussed in class.

a. Priority Inversion (W12C1)

b. Marshalling (W2C3)

c. Middleware (W3C2)

d. Stateless Server (W3C2)

e. Message Queue (W3C2)

f. Network Time Protocol (W3C3)

2. **(4 marks)** What is referential coupling? Identify or describe one of the distributed system architectures discussed in class that uses referential coupling, and one architecture that avoids referential coupling (W3C1).

3. **(8 marks)** Answer true or false by circling the correct answer.

In DDS (used in the 4th assignment), there is one queue per instance when the topic has a key.	TRUE	FALSE
When the Mars Pathfinder first landed, the bus was regularly reset because of a priority inversion.	TRUE	FALSE
A remote procedure call client stub must be statically configured to know the name and port of the RPC server.	TRUE	FALSE
When using persistent message communication, the message is lost if the receiver is not running at the time the message is sent.	TRUE	FALSE
Voting in decentralized mutual exclusion does not guarantee only one process will have access to a given resource.	TRUE	FALSE
Each entity has a unique access point.	TRUE	FALSE
In Raft elections the candidate with the longest log will be elected.	TRUE	FALSE
In a renewal frequency lease, if a replica pulls each time the lease expires, then the lease isn't long enough.	TRUE	FALSE

4. **(2 marks)** How was a content filtered topic used in the fourth assignment (A4)?

5. **(2 marks)** UDP networking is unreliable. What were the two types of errors that were simulated by the network library in the first two assignments (A1,A2)?

6. **(2 marks)** In what situation would you set a timeout on a socket (A1,A2,A3) when implementing a Remote Procedure Call (RPC)?

7. **(2 marks)** In real time systems, there are two general distributions used to characterize the behaviour of events/outputs/results. Identify or describe the two distributions.

8. **(4 marks)** We discussed four solutions to distributed mutual exclusion in class: token based, a central coordinator, Lamport clocks and Decentralized Voting. Describe how one of the last two methods (Lamport Clock or Decentralized Voting) works.

9. **(4 marks)** When scheduling real time systems, two processes may share a resource. If the processes are different priorities, the lower priority process may block the higher priority process from running. There are two scheduling algorithms designed to handle real time shared resource scheduling: Immediate Priority Ceiling Protocol (IPCP) and Original Priority Ceiling Protocol (OPCP). What are the differences between these two algorithms and give one advantage of each algorithm. (W12C1)

10. The Earliest Deadline First (EDF) scheduling algorithm is guaranteed to work if the utilization is less than 1. Consider the three periodic processes to be scheduled on a single CPU:

τ_i	C_i	T_i
1	1	8
2	2	5
3	3	9

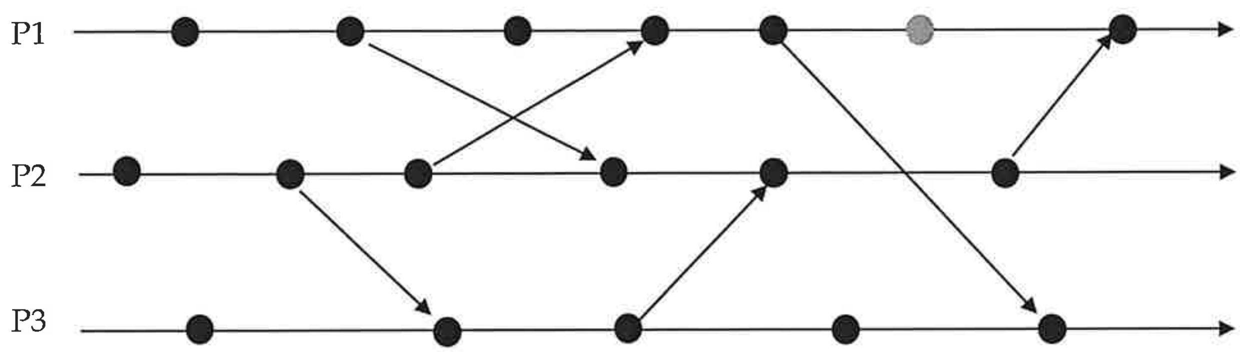
a. **(2 marks)** What is the total utilization of these three processes?(W11C3)

b. **(4 marks)** Provide a schedule for these process for the first 18 time units. Identify any periods of time where none of the processes are running (i.e. the CPU is idle). Use a vertical bar indicate a boundary between two processes or a boundary between a process and an idle period. (W11C3).

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

11. **(2 marks)** Two of the ways in which a system may fail are fail-stop and fail-arbitrary. What is the difference between the two?

12. The following diagram represents three processes with internal events and messages. Each black circle represents an event (W3C3).



a. **(4 marks)** Label the diagram using a vector clock expressed as a triple (three element tuple).
b. **(2 marks)** List ***all*** of the events (using their vector clock tuples) that are in the casual history of the grey circle (second from the right in P1).

12. **(3 marks)** Two of the election algorithms used to elect a coordinator that were discussed in class are the Bully and Ring algorithms. Describe how either one of these algorithms work (4C1/5C1-2).

13. **(10 marks)** The raft algorithm makes 5 guarantees, listed below. Provide a definition for each (W9C1).

a. Election Safety

b. Leader Append Only

c. Log Matching

d. Leader Completeness

e. State Machine Safety

14. **(2 marks)** Two approaches to consistency models are data centric consistency models and client consistency models. What is the difference between the two? (7C1)

14. **(2 marks)** The address of an entity in a distributed system may change. Two of the approaches in which the new address may be found are Forward Pointers and Home Based. Describe one of these two approaches. (W5C3)

15. In a distributed publish and subscribe system, requests by subscribers have to be matched with notifications by publishers, so that the data may be transferred.

a. **(2 marks)** What is the difference between topic based subscription and content based subscription? (W3C1)

b. **(2 marks)** Two methods of distributed event matching are Flooding and Selective Routing. Explain what a broker is, and how one of the two methods uses the brokers. (W5C1,2).

15. When dealing with replicated data, two general types of consistency conflicts are write-write and read-write conflicts (W6C2).

a. **(2 marks)** Define both types of conflicts.

b. **(2 marks)** How are quorums in vote based consistency protocols used to prevent each type of conflict?

16. When one copy of a replicated resource is updated, the three ways in which the other replicas can be updated is by propagating the notification, the data or the operation.

a. **(1 marks)** Which one is used by Raft?

b. **(2 marks)** Pick one of the methods and describe the characteristics of the data that would lead you to use it?

Extra Space.

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Extra Space.

Extra Space.